

Brewed Insights: From Raw Coffee Orders to an Interactive Excel Dashboard

1. Project Overview

This project analyzes a multi-year coffee sales dataset in Microsoft Excel and transforms raw transactional data into a structured, interactive dashboard. The objective is to combine spreadsheet-based data enrichment, formula logic, PivotTables, PivotCharts, formatting, a timeline, and slicers so that sales performance and customer preferences can be explored from a single business-facing view.

The workflow begins with three source sheets - orders, customers, and products. Customer and product attributes are brought into the orders data through lookup formulas, sales are calculated at line-item level, coded values are converted into readable category names, and the range is converted into an Excel Table. PivotTables then summarize sales over time, sales by country, and customer preference by coffee type. The finished dashboard adds interactive filtering by order date, roast type, package size, and loyalty-card status.

Project Component	Implementation
Primary Tool	Microsoft Excel
Source Sheets	orders, customers, products
Data Enrichment	XLOOKUP; INDEX-MATCH used during the build process; IF/IFS logic
Calculation	Sales = Unit Price x Quantity
Analysis	PivotTables and PivotCharts
Interactivity	Order Date timeline; Roast Type, Size, and Loyalty Card slicers
Dashboard Theme	Purple / lavender interactive sales and customer insights dashboard
Primary Analysis Themes	Sales trend, coffee preference, geography, roast, size, and loyalty-card filtering

2. Dataset and Analytical Model Summary

The source workbook contains separate transaction, customer, and product tables. The final analytical Orders table contains 1,000 sales line records and 16 fields after enrichment. Because an order can contain more than one product, Order ID is not a row-level unique key; the data contains 957 distinct order IDs across the 1,000 line items. The final data spans January 2, 2019 through August 19, 2022.

Dataset Characteristic	Value / Description
Sales line records	1,000
Distinct Order IDs	957

Dataset Characteristic	Value / Description
Customers in customer master	1,000
Customers represented in orders	913
Product variants	48
Sales period	January 2, 2019 - August 19, 2022
Total units sold	3,551
Total sales	\$45,134.26
Average sales per line	\$45.13
Countries	United States, Ireland, United Kingdom
Coffee types	Arabica, Liberica, Excelsa, Robusta
Roast types	Light, Medium, Dark
Package sizes	0.2 kg, 0.5 kg, 1.0 kg, 2.5 kg

A duplicate check found no exact duplicate rows in the final enriched dataset. Repeated Order IDs are legitimate multi-line orders rather than duplicate records: 31 order IDs occur on more than one product line, with a maximum of five lines for a single order. Missing email values are preserved as blanks rather than replaced with fabricated data; 206 order lines have no available customer email.

3. Data Preparation and Enrichment in Excel

The data-preparation stage converts the raw order sheet into a complete analytical table. The main transformation is lookup-based enrichment: Customer ID is used to retrieve customer attributes from the customers sheet, while Product ID is used to retrieve coffee attributes and pricing from the products sheet. This keeps the source tables normalized while giving the orders table all fields required for analysis.

Transformation	Excel Technique	Result
Customer Name	XLOOKUP	Adds the customer name to every order line.
Email	XLOOKUP wrapped in IF	Returns a blank when no email exists instead of displaying 0.
Country	XLOOKUP	Adds geographic information for country-level analysis.

Transformation	Excel Technique	Result
Coffee / Roast / Size / Unit Price	XLOOKUP	Adds product attributes and pricing from the products table.
Alternate lookup method	INDEX-MATCH	Used during the build workflow as an alternative lookup technique; the final saved workbook retains XLOOKUP formulas.
Sales	Multiplication formula	Calculates line sales as Unit Price x Quantity.
Readable category labels	IFS / multiple IF logic	Converts product codes such as Ara/Rob/Exc/Lib and L/M/D into full names.
Loyalty Card	XLOOKUP	Adds loyalty membership for dashboard filtering.

3.1 Lookup-Based Enrichment

The final workbook uses XLOOKUP extensively in the Orders table. For example, Customer Name is retrieved from the customers sheet by matching Customer ID, while product characteristics are retrieved by matching Product ID against the products sheet.

```
=XLOOKUP(C2,customers!$A$2:$A$1001,customers!$B$2:$B$1001)
```

For email addresses, the lookup is wrapped in an IF statement so that missing source values remain visually blank instead of being displayed as zero.

```
=IF(XLOOKUP(C2,customers!$A$2:$A$1001,customers!$C$2:$C$1001,,0)=0,"",XLOOKUP(C2,customers!$A$2:$A$1001,customers!$C$2:$C$1001,,0))
```

3.2 Sales Calculation and Category Mapping

Sales is calculated at the transaction-line level by multiplying quantity by unit price. This calculated field becomes the primary value used by all three PivotTables and dashboard charts.

```
=L2*E2
```

IFS logic converts abbreviated codes into readable labels. The coffee code mapping is Robusta, Excelsa, Arabica, and Liberica, while the roast mapping converts M, L, and D into Medium, Light, and Dark.

```
=IFS(I2="Rob","Robusta",I2="Exc","Excelsa",I2="Ara","Arabica",I2="Lib","Liberica")
```

```
=IFS(J2="M","Medium",J2="L","Light",J2="D","Dark")
```

4. Data Quality, Formatting, and Table Structure

After formula enrichment, the workbook applies date and number formatting so that dates, currency, and quantities are interpreted consistently. Duplicate checks are performed before analysis. The completed data range is then converted to an Excel Table named Orders, covering A1:P1001. Converting the range to a table provides structured references, consistent formatting, and a stable source for PivotTables and interactive controls.

Quality / Structure Step	Implementation	Purpose
Date formatting	Order Date formatted as a true Excel date	Supports chronological grouping, PivotTable

Quality / Structure Step	Implementation	Purpose
		date fields, and the timeline.
Number formatting	Currency and numeric formats applied	Improves readability of Unit Price and Sales.
Duplicate check	Checked before PivotTable construction	Avoids double-counting exact duplicate records.
Excel Table	Range A1:P1001 converted to table Orders	Creates a dynamic, structured analysis source.
Pivot source update	PivotTable data source refreshed / updated	Ensures newly added fields are available in the PivotTables.

5. PivotTable Analysis and Dashboard Construction

Three PivotTables provide the analytical foundation of the dashboard. The first summarizes Sum of Sales by year, month, and Coffee Type Name for the time-series view. The second summarizes Sum of Sales by Country. The third counts sales records by Coffee Type Name to represent customer preference. Each PivotTable is connected to PivotCharts and to common interactive controls.

Pivot Analysis	Configuration	Dashboard Visual
Total Sales Over Time	Rows: Year and Month of Order Date; Columns: Coffee Type Name; Values: Sum of Sales	Multi-series line chart
Sales by Country	Rows: Country; Values: Sum of Sales	Horizontal bar chart
Customer Preference by Coffee Type	Rows: Coffee Type Name; Values: Count of Sales	Horizontal bar chart

The workbook also contains a native Order Date timeline and slicers for Size, Roast Type Name, and Loyalty Card. These controls are connected to all three PivotTables, allowing the dashboard to update consistently when the user changes the analytical context.

5.1 Workflow Summary from the Build Process

Timestamp	Step	Role in the Project
00:00	Project Walkthrough	Introduces the workbook structure and final objective.
05:00	XLOOKUP	Populates customer and product attributes in the order data.
11:29	INDEX MATCH	Demonstrates an alternative lookup method during data enrichment.
17:46	Multiplication Formula for Sales	Creates the Sales field from quantity and unit

Timestamp	Step	Role in the Project
		price.
18:12	Multiple IF Functions	Converts abbreviated codes into readable categories.
21:15	Date Formatting	Prepares Order Date for time analysis.
22:50	Number Formatting	Formats monetary and numeric fields.
24:17	Check for Duplicates	Verifies data quality before aggregation.
24:45	Convert Range to Table	Creates the structured Orders table.
25:58	Pivot Tables and Pivot Charts + Formatting	Builds the analytical summaries and visuals.
32:05	Insert Timeline + Formatting	Adds interactive date filtering.
36:17	Insert Slicers + Formatting	Adds Roast Type, Size, and Loyalty Card filters.
36:48	Updating Pivot Table Data Source	Makes enriched fields available to the PivotTables.
49:22	Building the Dashboard	Combines charts and controls into the final layout.

6. Dashboard in Excel

The final dashboard, titled “Brewed Insights: Coffee Sales & Customer Trends,” combines the three analytical views with a consistent purple-and-lavender theme. The upper section contains the Order Date timeline and slicers, while the lower section is split between a large sales-trend chart and two supporting bar charts. The design keeps the primary time-series view dominant while allowing the user to compare preference and geography at a glance.

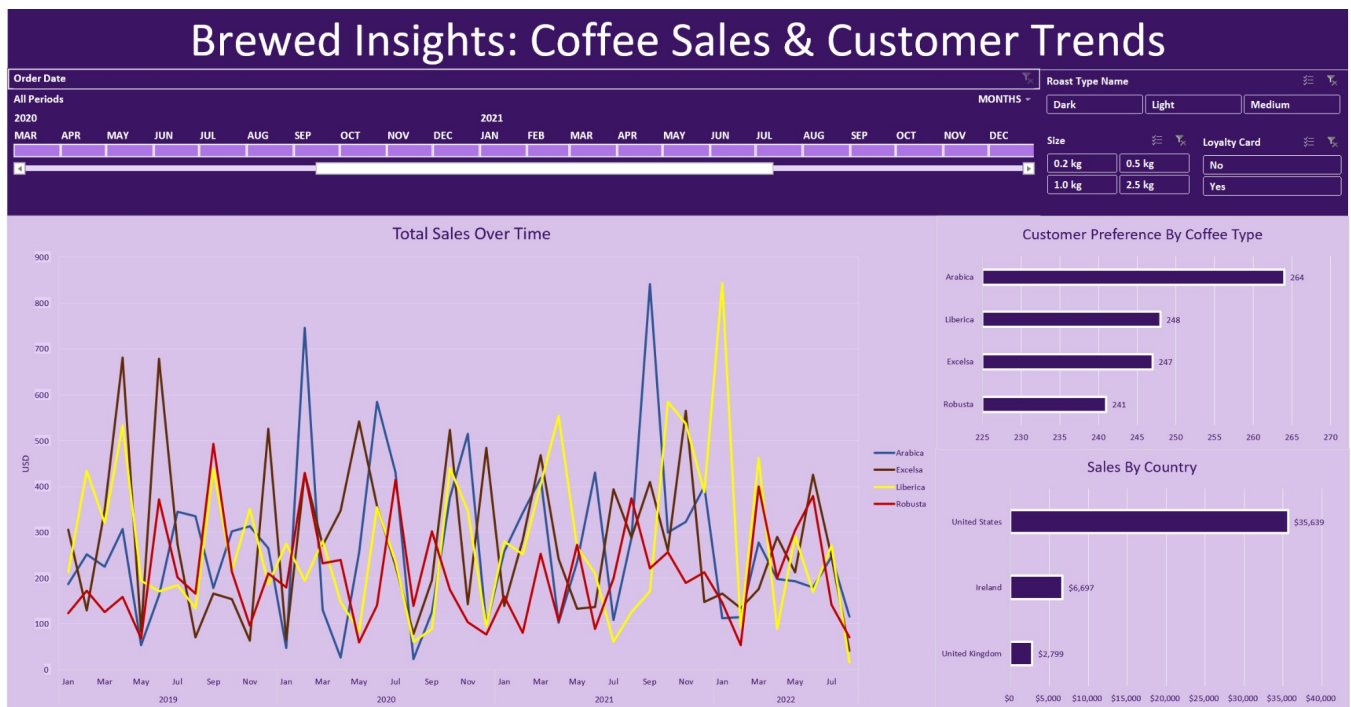


Figure 1. Brewed Insights: Coffee Sales & Customer Trends - interactive Excel dashboard with timeline, slicers, sales trend, preference, and country analysis.

6.1 Total Sales Over Time

The main line chart plots monthly sales from 2019 through August 2022 and separates the series by Arabica, Excelsa, Liberica, and Robusta. The monthly pattern is volatile rather than smooth, with different coffee types taking the lead in different periods. This view is best used to identify spikes, dips, and changing product mix over time rather than to infer a simple linear growth trend.

6.2 Customer Preference by Coffee Type

Customer preference is measured by the number of sales records for each coffee type. Arabica has the highest count at 264, followed by Liberica at 248, Excelsa at 247, and Robusta at 241. The narrow range between 241 and 264 indicates that customer preference is relatively balanced across the four coffee types.

6.3 Sales by Country

The geographic chart shows a strong concentration of sales in the United States. U.S. sales total \$35,638.88, compared with \$6,696.86 in Ireland and \$2,798.51 in the United Kingdom. This means the United States contributes approximately 79.0% of total sales in the dataset.

6.4 Interactive Filters

The dashboard can be filtered by Order Date, Roast Type Name, package Size, and Loyalty Card status. Because the timeline and slicers are connected to all PivotTables, a single selection changes the trend chart, customer-preference chart, and country chart together. This makes the workbook suitable for exploratory analysis without manually rebuilding formulas or charts.

7. Key Findings and Interpretations

- **Sales are geographically concentrated:** The United States contributes \$35,638.88, or about 79.0% of total sales, compared with 14.8% from Ireland and 6.2% from the United Kingdom.
- **Coffee preference is balanced by transaction count:** Arabica leads with 264 records, but the lowest category, Robusta, still has 241. No single coffee type dominates customer preference by count.
- **Revenue leadership differs from preference leadership:** Excelsa generates the highest sales at \$12,306.44 (27.3%), followed closely by Liberica at \$12,054.07 (26.7%) and Arabica at \$11,768.49 (26.1%). Robusta contributes \$9,005.25 (20.0%).

- **Light roast generates the most revenue:** Light roast sales total \$17,354.46 (38.5%), ahead of Medium at \$14,600.47 (32.3%) and Dark at \$13,179.32 (29.2%).
- **The 2.5 kg size is the highest-value format:** It produces \$23,785.56, or about 52.7% of total revenue, despite representing 247 of the 1,000 sales lines.
- **Loyalty-card customers are not the majority of revenue in this sample:** Orders linked to loyalty-card customers generate \$20,917.85 (46.3%) versus \$24,216.40 (53.7%) for non-card customers. The slicer allows this difference to be explored by product, roast, and period.
- **2022 is a partial year:** Sales total \$7,063.44 through August 19, 2022, so the 2022 total should not be compared directly with full-year 2019-2021 totals without adjusting for the shorter period.

8. Business Recommendations

- **Protect and expand the U.S. market while testing growth in Ireland and the U.K.:** The current revenue base is heavily concentrated in the United States, so international growth could diversify geographic risk.
- **Use coffee preference and revenue together:** Arabica has the most sales records, while Excelsa generates the most revenue. Product decisions should consider both customer frequency and monetary contribution rather than relying on one metric alone.
- **Prioritize the 2.5 kg segment in inventory and promotion planning:** Its revenue contribution is disproportionately high, making it an important format for stock availability and bundle decisions.
- **Investigate Light roast demand drivers:** Light roast is the highest-revenue roast category. Reviewing its performance by country, package size, and loyalty-card status may reveal the strongest combinations for targeted campaigns.
- **Use loyalty segmentation more actively:** The loyalty-card slicer can be extended into dedicated measures such as average sales per customer, repeat purchase rate, and product mix to determine whether the program is creating incremental value.
- **Track seasonality with full-year comparisons:** The monthly chart shows substantial variation. Future analysis should compare like-for-like months and complete years before drawing conclusions about long-term growth or decline.

9. Conclusion and Future Scope

This project demonstrates an end-to-end Excel analytics workflow that begins with separate raw transaction, customer, and product tables and ends with a polished, interactive dashboard. The technical core of the project is the progression from lookup-based enrichment and formula-driven calculations to an Excel Table, PivotTables, PivotCharts, a timeline, slicers, and a consolidated dashboard. The final workbook keeps the analysis accessible to a business user while preserving traceable spreadsheet logic behind each visual.

The strongest practical value of the project is its integration of data preparation and visualization inside one Excel workbook. XLOOKUP and IF/IFS logic turn the raw transaction table into an analysis-ready dataset; PivotTables aggregate that data efficiently; and the timeline and slicers convert static charts into an exploratory tool. The dashboard makes it possible to move from a high-level sales trend to focused questions about roast type, size, loyalty status, coffee preference, and country without changing the underlying model.

Future work can extend the workbook with KPI cards for total sales, order count, average order value, and units sold; customer-level repeat-purchase analysis; profit and margin metrics; monthly and year-over-year growth measures; dedicated loyalty-program comparisons; top-product rankings; and dynamic named measures. A more advanced version could also use Power Query for repeatable data cleaning and automated refresh, Power Pivot / the Data Model for scalable relationships, and additional drill-down views for customer and product performance.